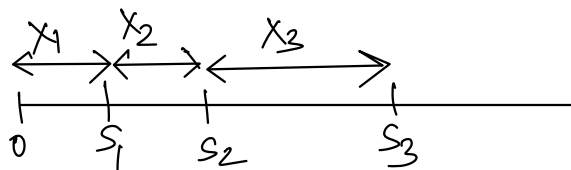


We have a Poisson process with rate λ

$\Rightarrow$ the inter-arrival times $X_1, X_2, \dots \stackrel{iid}{\sim} \exp(\lambda)$

To prove: the counting process $N_t \sim \text{Poi}(\lambda t)$

1) Find joint density of $X_1, X_2, \dots, X_n$.



Example: X_1, X_2, X_3

$$\text{Joint pdf: } f_{X_1, X_2, X_3}(x_1, x_2, x_3) = \begin{cases} \lambda e^{-\lambda x_1} (\lambda e^{-\lambda x_2}) (\lambda e^{-\lambda x_3}) & \text{if } x_1 > 0, x_2 > 0, x_3 > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore f_{X_1, X_2, X_3}(x_1, x_2, x_3) = \begin{cases} \lambda^3 e^{-\lambda(x_1 + x_2 + x_3)} & , (x_1, x_2, x_3) \in (\mathbb{R}^+)^3 \\ 0 & , \text{otherwise} \end{cases}$$

Similarly

$$f_{X_1, X_2, \dots, X_n}(x_1, \dots, x_n) = \begin{cases} \lambda^n e^{-\lambda(x_1 + x_2 + \dots + x_n)} & , (x_1, \dots, x_n) \in (\mathbb{R}^+)^n \\ 0 & , \text{otherwise} \end{cases}$$

2) Find joint density of the arrival epochs $S_1, \dots, S_n$ using the joint density of $X_1, \dots, X_n$

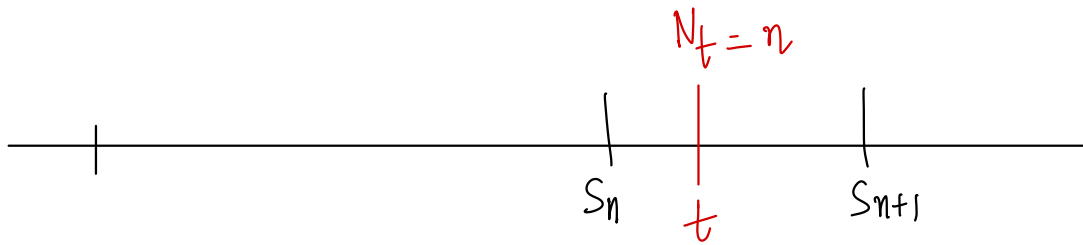
$$\begin{array}{l|l} \begin{array}{l} S_1 = X_1 \\ S_2 = X_1 + X_2 \\ S_3 = X_1 + X_2 + X_3 \\ \vdots \\ S_n = X_1 + \dots + X_n \end{array} & \begin{array}{l} \text{Suppose} \\ n=3 \end{array} \end{array} \quad \left| \frac{\partial \underline{S}}{\partial \underline{X}} \right| = \begin{vmatrix} \frac{\partial S_1}{\partial X_1} & \frac{\partial S_1}{\partial X_2} & \frac{\partial S_1}{\partial X_3} \\ \frac{\partial S_2}{\partial X_1} & \frac{\partial S_2}{\partial X_2} & \frac{\partial S_2}{\partial X_3} \\ \frac{\partial S_3}{\partial X_1} & \frac{\partial S_3}{\partial X_2} & \frac{\partial S_3}{\partial X_3} \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = \underline{\underline{1}}$$

$$f_{S_1, S_2, S_3}(t_1, t_2, t_3) = \begin{cases} \lambda^3 e^{-\lambda t_3} & , 0 < t_1 < t_2 < t_3 \\ 0 & , \text{otherwise} \end{cases}$$

$$\text{Similarly, } f_{S_1, \dots, S_n}(t_1, t_2, \dots, t_n) = \begin{cases} \lambda^n e^{-\lambda t_n} & , 0 < t_1 < \dots < t_n \\ 0 & , \text{otherwise} \end{cases}$$

3) Required: find $P\{N_t = n\}$

$$\{N_t = n\} \text{ is same as } \left\{ (t_1, t_2, \dots, t_n, t_{n+1}) \in (\mathbb{R}^+)^{n+1} \text{ s.t. } 0 < t_1 < t_2 < \dots < t_n \leq t < t_{n+1} \right\}$$



$$S_1 = t_1, S_2 = t_2, \dots, S_n = t_n, S_{n+1} = t_{n+1}$$

$$A_{n,t} = \left\{ (t_1, t_2, \dots, t_n, t_{n+1}) \text{ s.t. } 0 < t_1 < t_2 < \dots < t_n \leq t < t_{n+1} \right\} \\ = \{N_t = n\}$$

$$f_{S_1, \dots, S_{n+1}}(t_1, t_2, \dots, t_{n+1}) = \begin{cases} \frac{\lambda^{n+1} e^{-\lambda t_{n+1}}}{0}, & 0 < t_1 < t_2 < \dots < t_{n+1} \\ 0, & \text{otherwise} \end{cases}$$

$$f_{S_1, \dots, S_{n+1}} \Big|_{(N_t = n)}(t_1, t_2, \dots, t_{n+1}) > 0$$

$$(t_1, t_2, \dots, t_{n+1}) \in (\mathbb{R}^+)^{n+1} \\ 0 < t_1 < t_2 < \dots < t_{n+1}$$

$$(t_1, t_2, \dots, t_{n+1}) \in (\mathbb{R}^+)^{n+1}$$

$$\text{s.t. } 0 < t_1 < t_2 < \dots < t_n \leq t < t_{n+1}$$

$$\underbrace{f_{S_1, \dots, S_{n+1}} \Big|_{N_t = n}}_{\text{conditional pdf}}(t_1, t_2, \dots, t_{n+1} \mid N_t = n) = \begin{cases} \frac{\lambda^{n+1} e^{-\lambda t_{n+1}}}{P(N_t = n)} & \text{for all those} \\ & (t_1, t_2, \dots, t_{n+1}) \in A_{n,t} \\ 0 & \text{elsewhere} \end{cases}$$

$$f_{S_1, \dots, S_n} \Big|_{N_t = n} = \int_t^\infty f_{S_1, \dots, S_{n+1}} \Big|_{N_t = n} dt_{n+1} \\ = \int_t^\infty \frac{\lambda^{n+1} e^{-\lambda t_{n+1}}}{P(N_t = n)} dt_{n+1}$$

$$= \frac{1}{P(N_t=n)} \lambda^n \int_t^{\infty} \lambda e^{-\lambda t_{n+1}} dt_{n+1}$$

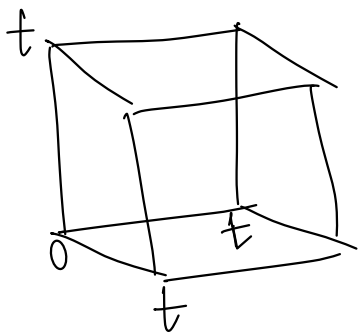
$$f_{S_1, \dots, S_n} | N_t = n = \frac{1}{P(N_t=n)} \lambda^n e^{-\lambda t}$$

$$f_{S_1, \dots, S_n} | N_t = n (S_1 = t_1, S_2 = t_2, \dots, S_n = t_n) = \frac{1}{P(N_t=n)} \lambda^n e^{-\lambda t}$$

for all $(t_1, \dots, t_n) \in (\mathbb{R}^+)^n$
 s.t. $0 < t_1 < t_2 \dots < t_n \leq t$

$$f_{S_1, S_2, S_3} | N_t = 3 (S_1 = t_1, S_2 = t_2, S_3 = t_3) = \begin{cases} \frac{1}{P(N_t=3)} \lambda^3 e^{-\lambda t} & , 0 < t_1 < t_2 < t_3 \leq t \\ 0 & \text{otherwise} \end{cases}$$

$$(\mathbb{R}^+)^3 \quad \underline{(t_1, t_2, t_3) \in (\mathbb{R}^+)^3} \text{ s.t. } 0 \leq t_1 \leq t, 0 \leq t_2 \leq t, 0 \leq t_3 \leq t$$



$$(t_1, t_2, t_3) \quad t_1 < t_2 < t_3$$

$$\text{if } t = 2$$

$$(1, 2.5, 0.5) \quad t_1 < t_2 < t_3$$

$$(t_1, t_2, t_3)$$

$$\boxed{t_1 < t_2 < t_3}$$

$$t_1 < t_3 < t_2$$

$$t_2 < t_1 < t_3$$

$$t_2 < t_3 < t_1$$

$$t_3 < t_1 < t_2$$

$$t_3 < t_2 < t_1$$

$$\frac{t^3}{3!}$$

Volume:

$$\{(t_1, \dots, t_n) \in (\mathbb{R}^+)^n \text{ s.t. } 0 < t_1 < t_2 < \dots < t_n \leq t\}$$

$$\frac{t^n}{n!}$$

$$f_{S_1, \dots, S_n} | N_t = n = \frac{1}{\text{volume of this set}} = \frac{t^n}{n!}$$

$$\frac{\lambda^n \cdot e^{-\lambda t}}{P\{N_t = n\}} = \frac{n!}{t^n} \Rightarrow P\{N_t = n\} = \frac{(\lambda t)^n e^{-\lambda t}}{n!} = \text{Pos}(\lambda t)$$

$$N_t \sim \text{Pos}(\lambda t)$$